Semantic-pragmatic Constraints On Chinese Genitive Relative Construction

Introduction

This study investigates Chinese genitive relative constructions (e.g. $N_{possessee}$+predicate+genitive marker+$N_{possessor}$) to argue that grammatical acceptability depends not only on syntactic well-formedness, but also on semantic and pragmatic constraints governing reference-point selection. Through contrastive analysis of adjectival and verbal predicates, I show that adjectival genitive relative constructions require $N_{possessee}$ to exhibit low individuation and high conceptual dependency on $N_{possessor}$, functioning as an inherent attribute. Verbal genitive relative constructions, though syntactically more flexible, are pragmatically restricted to tracking highly accessible, context-anchored events rather than introducing new information. I demonstrate that gradient acceptability is systematically regulated by cognitive principles such as salience and accessibility. The study offers a unified account of genitive relative syntax that complements generative operations with cognitive constraints, contributing to cross-linguistic theories of the semantics-pragmatics interface and the role of cognitive construal in grammatical structure.

Keywords: genitive relative constructions; semantic-pragmatic constraints; clausal relativization

1. Introduction

As a useful diagnostic reference (Yuan 1994; Chi et al. 2022), a monovalent noun requires a semantically dependent nominal element for full interpretation, such as kinship, part-whole, and attribute nouns. Such nouns can't be fully interpreted without a possessor, and may appear in the pre-de subject position of genitive relative constructions, as illustrated in (1). In (1), the monovalent noun "volume" functions as $N_{possessee}$, while the head noun "container" serves as $N_{possessor}$. The empty category *e* is activated by this monovalent noun. *e* is co-indexed with the head "container", yielding the reading "the container's volume is large".

(1) 体积 e_i 很 大 的 容器$_i$

tǐ jī hěn dà de róng qì
volume very large GEN container
'a container with a large volume'

(1) instantiates one subtype of the broader empirical target of this study, the genitive relative construction. Before proceeding further, it is critical to demarcate the two major subtypes examined throughout the analysis, since both build on 的-marked relative clause modification yet diverge in predicate makeup and interpretive behaviour. The adjectival-predicate genitive relative construction follows the abstract template $N_{possessee}$+Adj+GEN +$N_{possessor}$. In this pattern, the pre-的 clausal subject corresponds semantically to $N_{possessee}$, the adjective supplies a stable state-level predicate, and the final head noun functions as $N_{possessor}$ that bears a plausible possessive-style semantic link to the pre-的 subject (e.g. 屏幕精致的电视 'the television with a delicate screen'; see (25a)). By contrast, the verbal-predicate genitive relative construction obeys the template $N_{possessee}$+VP+GEN+$N_{possessor}$. Here, the pre-的 subject is again $N_{possessee}$, but the predicate is an event-denoting verbal phrase rather than a stative adjective (e.g. 数据没上传的员工 'the employee whose data are not uploaded'; see (35)).

It is important to stress from the outset that while monovalent nouns frequently occupy the $N_{possessee}$ slot as seen in (1), neither subtype is restricted exclusively to monovalent $N_{possessee}$. Non-monovalent nouns may also fill this position when suitable semantic-pragmatic conditions hold, functioning as clausal subjects to establish a syntax-semantics link with their heads, as illustrated in (2a). This structure is also well-known as gapless relative clause (Lee & Lee, 2012; Li & Kang, 2024; Pan, 2022; Tsai, 2008; Wen, 2020). The phrase 动物繁多 'many animals' enclosed in brackets does not have an overt gap coindexed with the head 农场 'farm'. In contrast, a gapped or standard relative clause has a present gap co-indexed with the head 书 'book', instantiated by (3).[1]

(2a) [动物 繁多] 的 农场
dòng wù fán duō de nóng chǎng
animal many GEN farm
'a farm with many animals'

[1] For interested readers, regarding further theoretical discussion about gapless relative clauses, see complex NP analysis (Murasugi, 1991; Tsai, 2008), conjunction analysis (Loock, 2010), adjunct analysis (Ning, 1993), and preposition ghosting analysis (Collins & Radford, 2015).

(2b) 农场 里 的 动物 繁多

nóng chǎng lǐ de dòng wù fán duō

farm in GEN animal many

'There are many animals on the farm.'

(3) 比尔 看了 e_i 的 书$_i$ 繁多

bǐ ěr kàn le de shū fán duō

Bill read-ASP GEN book many

'Bill read many books.'

(Li & Kang, 2024)

While some non-monovalent nouns can be relativized when they act as subjects, most are still very restricted in the relativization process. (4a) is not allowed to be transformed into (4b)[2]. The comparison reflects that subtle differences in the genitive relation between subject and head can affect the grammaticality of an entire construction.

(4a) 这个 先生 的 手机 很 好看

zhè gè xiān shēng de shǒu jī hěn hǎo kàn

this gentleman GEN phone very good-looking

'This gentleman's phone is very good-looking.'

(4b) *手机 很[3] 好看 的 先生

shǒu jī hěn hǎo kàn de xiān shēng

[2] As discussed in section 5, while 手机 'cell phone' satisfies the constraint of inanimacy and arguably exhibits a high degree of pragmatic dependency in modern life, the construction still remains infelicitous. This issue reveals that the proposed constraints are not sufficient in isolation. The key lies in the interplay between $N_{possessee}$ and the predicate within a specific cognitive frame. "Good-looking" is an aesthetic property that primarily characterizes the phone itself, not the gentleman. There is no conventional or contextually salient frame in which the aesthetic quality of a man's phone serves as a defining or characteristic property of the man himself. In other words, the predicate-possessee combination ("good-looking phone") fails to be construed as an intrinsic or revealing property about $N_{possessor}$. The crucial factor of grammaticality is not entirely the dependency of $N_{possessee}$ per se, but also whether the predicate-possessee combination can achieve interpretive relevance, namely whether it can be understood as expressing a cognitively central or diagnostically revealing property of $N_{possessor}$ within a plausible cognitive frame. This relevance is dynamically conditioned by world knowledge and contextual salience, which accounts for the gradient and sometimes context-sensitive nature of acceptability judgments.

[3] The unacceptability of (4b) can't be reduced to the absence of a degree adverb, a general requirement for adjectival predicates in Mandarin. Even with 很 'very' supplied, 手机很好看的先生 'the gentleman with a very good-looking phone' remains substantially less natural than syntactically parallel cases like 屏幕很精致的电视 'the television with a very delicate screen'. The residual contrast persists because a good-looking phone, unlike a delicate screen, is not conventionally construed as a defining property of its possessor, the cognitive-pragmatic constraint that constitutes my central claim.

phone very good-looking GEN gentleman

'the gentleman with a very good-looking phone'

To preview, the objective of this study is to investigate the semantic-pragmatic constraints governing two types of Chinese genitive constructions: those with adjectival predicates ($N_{possessee}$+Adj+GEN+$N_{possessor}$) and those with verbal predicates ($N_{possessee}$+VP+GEN+$N_{possessor}$). I offer a novel analysis to contribute to a more integrated understanding of how cognitive, semantic, and pragmatic factors interact in the formation of Chinese genitive structures.

2. Generative Framework

I start by reviewing how the genitive relation is defined specifically in Chinese under the generative framework. Genitive relative constructions usually occur in the form of $N_{possessor}$ 的 $N_{possessee}$, where a DP that precedes 的 denotes $N_{possessor}$ (Cheng, 1986). $N_{possessee}$ of $N_{possessor}$ in English corresponds to $N_{possessor}$ 的 $N_{possessee}$ in Chinese. The morpheme 的 functions as the head of the maximal projection DP. Research has shown the heterogeneity of XPs preceding 的 (e.g. Paul, 2017; see 5a-5f).

(5a) [$_{DP}$ 玛丽] 的 电脑

mǎ lì de diàn nǎo

Mary GEN computer

'Mary's computer'

(5b) [$_{NP}$ 黑头发] 的 女孩

hēi tóu fǎ de nǚ hái

black hair GEN girl

'the girl with black hair'

(5c) 三个 [$_{QP}$ 五岁] 的 孩子

sān ge wǔ suì de hái zi

3-CL 5 year GEN child

‘three five-year-old children’

(5d) [PP 对他] 的 印象
duì tā de yìn xiàng
of him GEN impression
‘impression of him’

(5e) 一个 [AdjP 方] 的 盘子
yí ge fāng de pán zi
1-CL square GEN plate
‘a square plate’

(5f) [DP [TP 你 寄来] 的 信]
nǐ jì lái de xìn
2SG send GEN letter
‘the letter you sent’

Wen’s PhD dissertation (2020) examines the syntax of genitive relative constructions in Mandarin Chinese, focusing on adjunct and gapless clauses. The study argues against the traditional null operator movement analysis and proposes a head-raising strategy to derive these constructions, providing a better explanation for their properties and interactions with comparative deletion. For adjunct relative clauses, the study distinguishes between those with gaps and those with resumptive pronouns. Adjunct relatives with gaps (e.g. locative, time, instrumental, manner, and reason relatives) are analyzed by using the head-raising strategy, where the head noun raises from within the relative clause. For example, in (6), the head noun 大学 ‘university’ is raised from within the clause. In contrast, adjunct relatives with resumptive pronouns (e.g. comitative, goal, and ablative relatives) are analyzed via base generation, as movement would violate the ban on preposition stranding in Mandarin. For instance, in (7), the resumptive pronoun 她 ‘her’ is base-generated. Gapless relatives, on the other hand, are true relatives rather than noun complements. They are divided into two types: adjunct gapless relatives (see (8)) and resultative gapless relatives (see (9)). The former involves a manner-kind head noun, while the latter involves a result-kind head noun

with a null VP.

(6) 这 就 是 那 所 Lisa 学 了 4 年 语言学 的 大学
zhè jiù shì nà suǒ Lisa xué le 4 nián yǔ yán xué de dàxué
this exactly COP that CL Lisa study PFV 4 year linguistics GEN university
'This is the university where Lisa studied linguistics for 4 years.'

(7) *这 就 是 我 跟 她 跳舞 的 姑娘
zhè jiù shì wǒ gēn tā tiàowǔ de gūniang
this exactly COP 1SG with 3SG dance GEN girl
'This is the girl (with whom) I danced.'

(8) 这 是 Lisa 弹 钢琴 的 姿势
zhè shì Lisa tán gāngqín de zī shì
this COP Lisa play piano GEN posture
'This is the posture that Lisa had when playing piano.'

(9) 这 是 Lisa 弹 钢琴 的 下场
zhè shì Lisa tán gāngqín de xià chǎng
this COP Lisa play piano GEN consequence
'This is the consequence that Lisa had when playing piano.'

Wen's (2020) syntactic distinction aligns with Lin's (2018) argument, which contrasts two kinds of *的*-marked relative-clause structures. Specifically, Wen (2020) differentiates adjunct-type gapless relatives from possessive-link gapless relatives; Lin's (2018) test materials map exactly onto these two syntactic classes. One group is the adjunct-type gapless relative exemplified in (8). The other group covers constructions where the head noun establishes a possessive link to one nominal inside the pre-*的* clause, illustrated by (36). The experimental data show that when the possessive-linked nominal sits in the embedded subject position, readers can resolve the head-clause-internal nominal dependency more easily.

While Wen (2020) approaches gapless relatives through head-raising and distinguishes adjunct from resultative types, Li and Kang (2024) pursue a fundamentally different direction: they argue that

these constructions are not gapless at all, but contain a covert gap created by the deletion of a preposition or a verb. Regarding the PP-type, in a phrase like 修车的车库 'the garage where (someone) fixes the car', the locative preposition 在 'in' is absent from the surface string. 车库 'garage' undergoes relativization. Because Mandarin independently bans preposition stranding, the preposition is forced to pied-pipe with its complement into [Spec, PP] before reaching [Spec, CP]. This creates a configuration where both the head P and its specifier would be spelled out overtly, violating Collins' (2007) edge constraint. The preposition is therefore ghosted at phonological form via the preposition ghosting operation (Collins & Radford, 2015), yielding the surface appearance of gaplessness. Critically, not all prepositions can be ghosted. Li and Kang (2024) propose an inference hierarchy: prepositions encoding place, time, manner, instrument, and reason are sufficiently inferable from the head noun to permit deletion, whereas those encoding comitative, goal, dative, ablative, and comparative roles are not, and must surface with a resumptive pronoun, as in 我跟他跳舞的姑娘 'the girl with whom I danced'.[4] These derivational accounts supply two concrete diagnostic criteria: (i) how to verify whether the nominal inside the pre-的 clause functions syntactically as an embedded subject; (ii) how to tell possessive link interpretations apart from pure event modification readings. These diagnostics are relevant here because they help identify when a pre-的 nominal is genuinely construed as an embedded subject bearing a possessive link to the head, rather than as part of an event-modification or gapless reading. This is a key distinction that underlies the acceptability contrasts discussed from section 5 to 6.

The VP-type operates through serial verb constructions. In 卖马(挣)的钱 'the money earned by selling the horse', the verb 挣 'earn' is missing. Li and Kang (2024) analyze such cases as involving two VPs in a cause-effect relation---here, a resultative one. When the complement of the second verb undergoes relativization, the verb itself is deleted but remains recoverable through coercion: the head noun 钱 'money' coerces the agentive reading 'earn' because the two VPs form a single event in the sense of Truswell's (2011) single event grouping condition, which attenuates the adjunct island effect and licenses extraction. The same mechanism applies to instrumental serial verb constructions such as 买车的钱 'the money used to buy a car', where the first verb 用 'use'

[4] It should be noted that canonical adjunct-type gapless relatives, such as (8), are semantically distinct from my target genitive relative constructions from section 5. In adjunct-type cases, the head noun takes the whole pre-的 clause purely as an event argument, rather than establishing a possessive-style link to one nominal inside the modifying clause. I survey work on gapless relative structures nevertheless, because key generative analyses of genitive relative constructions (including Lin, 2018; Wen, 2020) build their arguments by drawing explicit contrasts against adjunct-type gapless relatives. To properly evaluate their claims about genitive relative constructions, this contrasting baseline must be introduced.

is omitted and recovered via coercion from the head noun.

Ke's (2023) analysis of Chinese genitive topic constructions based upon the nature of null arguments in different NPs provides a powerful explanation for their syntactic behaviors. For kinship nouns, consider (10). The null possessor of the kinship noun 儿子 'son' is a long-distance reflexive argument. Kinship nouns bear a null long-distance reflexive genitive argument that seeks an antecedent, and due to its subject orientation, it prefers to take the subject as $N_{possessor}$. In this case, if we assume the topic is 张三 'Zhangsan' and the subject within the clause is 李四 'Lisi'[5], the kinship noun would prefer to take 李四 as its possessor.

(10) 张三 在 李四 住院 时 带 儿子 去 了 美国
Zhāng Sān zài Lǐ Sì zhùyuàn shí dài érzi qù le měiguó
Zhangsan at Lisi hospitalize when take son go PFV America
'Zhangsan$_i$, during the time when Lisi$_j$ was hospitalized, took his$_{i/*j}$ son to America.'

For body-part nouns, consider (11). The null argument of the body-part noun 手 'hand' is a locally bound reflexive, bound by the local subject 李四 'Lisi', making it difficult for the topic 张三 'Zhangsan' to be related to the comment. However, when a body-part noun is in the embedded subject position, as in (12), its binding domain can be extended to the superordinate clause, allowing the topic to be its possessor.

(11) 张三 意识 到 李四 快 弄 伤 手 了
Zhāng Sān yìshi dào Lǐ Sì kuài nòng shāng shǒu le
Zhangsan realize arrive Lisi soon do injure hand PFV
'Zhangsan$_i$ realized that Lisi$_j$ was about to hurt his$_{*i/j}$ hand.'

(12) 张三 呢 李四 听说 今天 脚 受 伤 了
Zhāng Sān ne Lǐ Sì tīngshuō jīntiān jiǎo shòu shāng le
Zhangsan TOP Lisi hear of today foot suffer injure PFV
'For Zhangsan$_i$, Lisi$_j$ hears that he$_{i/*j}$ hurt his foot today.'

For whole-part nouns with an inanimate possessor, consider (13) as the topic construction.

[5] Lisi is a gender-neutral Chinese proper name. The gloss uses *his* solely as an English placeholder for the indexed possessor$_i$. It imparts no masculine gender to Lisi in the source Chinese sentence. (11) follows the same convention.

Whole-part nouns like 封面 'cover' bear a pronominal genitive argument. This rule allows a whole-part noun to be related to the topic relatively freely, as here, a cover usually belongs to a book without being restricted by the same syntactic conditions as kinship nouns and body-part nouns. A whole-part noun with an inanimate possessor can occur in various syntactic positions and still maintain a proper relation with the topic, which is in contrast to the behaviors of kinship nouns and body-part nouns and is well-explained by the pronominal nature of its null argument.

(13) 这 本 书 呢， 封面 是 日本 画家 设计 的
zhè běn shū ne fēng miàn shì rìběn huàjiā shèjì de
this CL book TOP cover COP Japan painter design GEN
'For this book, the cover was designed by a Japanese artist.'

3. Cognitive Framework

Much research semantically categorizes the genitive relation with inconsistent approaches, as the number of categories ranges from three to more than two dozen (e.g. Lu, 2002; Dixon, 2010). Although it is difficult to define the connotation and denotation of the genitive relation (Heine 1997), various studies generally agree that the genitive relation has archetypal characteristics. Liao (1986) proposes that when B is a(n) component/property/aspect of A or shares a(n) state/entity/event with A, A provides B with a cognitive framework for further disaggregating A or associating it with B, allowing the genitive relation. The present study adopts this broad interpretation.

It is also important to determine whether $N_{location}$ of N_{entity} contains $N_{possessee}$ of $N_{possessor}$ for the following discussion. Lyons (1967), Clark (1978), Croft (1990), and Hu (2009) all agree that the categories of existence, location, and genitive relation are closely related, historically deriving from the category of location. The possessor of the genitive category is animate, and possession is the subcategorization of the locational category. In the following discussion, we can also find that the representation of the locational category is very similar to that of the genitive category with consistent analysis. Therefore, I treat $N_{location}$ of N_{entity} as $N_{possessee}$ of $N_{possessor}$.

Since Zhu (1961, 1978), it has long been proposed that 的 can be used in genitive structures to

indicate the genitive relation. Previous research also uncovers under what conditions 的 is allowed (Si, 2017). By comparing alienable and inalienable genitive structures, it is found that 的 is mandatory in an alienable genitive phrase (where $N_{possessee}$ can be separated or removed from $N_{possessor}$) (see (14)), but is optional in an inalienable phrase (where $N_{possessee}$ can't be separated or removed from $N_{possessor}$) (see 15a-15c).

(14) 我 的 书包
wǒ de shū bāo
1SG GEN bag
'my bag'

(15a) kinship
我 (的) 父亲
wǒ (de) fù qīn
1SG (GEN) father
'my father'

(15b) part-whole relation
车 (的) 轮胎
chē (de) lún tāi
car (GEN) tyre
'car tyre'

(15c) locative relation
食堂 (的) 后面
shí táng (de) hòu miàn
canteen (GEN) back
'the back of the canteen'

Secondly, copula transformation does not read well in inalienable genitive structures (16a→16b), but becomes felicitous in alienable genitive structures (17a→17b) (Si, 2017).

(16a) 简 的 祖父
jiǎn de zǔ fù
Jane GEN grandfather

‘Jane’s grandfather’

(16b) *祖父 是 简 的
zǔ fù shì jiǎn de
grandfather COP Jane GEN
*‘Grandfather is Jane’s.’

(17a) 我 的 海报
wǒ de hǎi bào
1SG GEN poster
‘my poster’

(17b) 海报 是 我 的
hǎi bào shì wǒ de
poster COP 1SG GEN
‘The poster is mine.’

Thirdly, ditransitive transformation tends to produce marked, dispreferred readings for inalienable possession readings (18a→18b)[6], but becomes natural in alienable genitive structures (19a→19b) (Si, 2017).

(18a) 我 的 腿
wǒ de tuǐ
1SG GEN leg
‘my leg’

(18b) ? 我 给了 你 一条 腿[7]
wǒ gěi le nǐ yì tiáo tuǐ
1SG give-PST 2SG one-CL leg

[6] Well-formed counter-examples exist when the verb encodes symbolic or ritual action without actual transfer of the inalienable bond. For instance, 他献给河神一个儿子 ‘He offered a son to the River God.’ is acceptable under a sacrificial reading. The verb encodes a ritual offering rather than real transfer of the kin-based inalienable bond between father and son.

[7] The sentence may work when 腿 ‘leg’ does not belong to any human being, but rather refer to a piece of meat (e.g. chicken), such as 我给了你一条腿，你却什么也没给我 ‘I gave you one (chicken) leg, but you gave me nothing’.

‘I gave you a leg’

(19a) 我 的 杯子

wǒ de bēi zi

1SG GEN cup

‘my cup’

(19b) 我 给了 你 一个 杯子

wǒ gěi le nǐ yí gè bēi zǐ

1SG give-PST 2SG one-CL cup

‘I gave you a cup.’

Lastly, possessee omission is ungrammatical in inalienable genitive structures (20a→20b), but becomes acceptable in alienable genitive structures given an appropriate context (21a→21b) (Si, 2017).

(20a) context

我 喜欢 这本 书 的 封面

wǒ xǐ huān zhè běn shū de fēng miàn

1SG like this-CL book GEN cover

‘I like the cover of this book.’

(20b) possessee omission

*我 喜欢 这本[8] 书 的

wǒ xǐ huān zhè běn shū de

1SG like this-CL book GEN

*‘I like the book’s.’

(21a) context

[8] I argue that the ungrammaticality of possessee omission in (12b) may not be absolute but contingent on referential accessibility. Replacing the proximal demonstrative 这本 ‘this’ with the distal 那本 ‘that’ can improve acceptability because the latter typically signals a referent already established in the shared discourse context. In such a case, the omitted possessee 封面 ‘cover’ becomes highly accessible and recoverable, mitigating the infelicity. This phenomenon suggests that the constraint on possessee omission in inalienable genitives is not merely a semantic (inalienability) rule but is pragmatically modulated by the discourse prominence and recoverability of the referent, aligning with the broader cognitive-functional principle of accessibility governing syntactic reduction.

这里　有　三双　手套

zhè lǐ　yǒu　sān-shuāng　shǒu-tào

here　have　three-CL　glove

‘Here are three pairs of gloves.’

(21b) possessee omission

我　能　拿　我　的　吗

wǒ　néng　ná　wǒ　de　ma

1SG　cab　take　1SG　GEN　Q

‘Can I take mine?’

A recent constructional account by Zhang & Wang (2025) offers a semantic taxonomy of the N_1 的 N_2 structure based upon corpus data. They identify three main constructional subtypes. Performative interaction possession covers cases like *my clothes*, where the possessor actively controls the possessee. Intersubjective possession covers *my wife* and *his friend*, where the relation is reciprocal and status-symmetric. Subjectively construing possession covers a wider range, including part-whole (e.g. the table’s legs), location (e.g. the athletes in the playground), and material (e.g. the wooden table). Their framework maps these into a schematic network organized by prototype and schema. What their analysis achieves is a systematic inventory of what meanings the N_1 的 N_2 structure can encode. However, it does not explain why some configurations, though semantically classifiable, are judged less acceptable than others, nor why minimal contextual shifts can upgrade or downgrade acceptability.

4. The Present Study

Previous research under the generative framework has discussed the minimal projection of 的 (XP 的 NP), the structure of genitive constructions among all 的-related structures, and the conditions when 的 is optional and required, and the syntax of genitive constructions in Mandarin Chinese (particularly adjunct and gapless clauses). The generative framework offers a precise, structural

skeleton[9]: it defines the possible syntactic configurations (e.g. the DP hypothesis for 的, the head-raising analysis for relative clauses) and identifies the grammaticality boundaries of the construction. However, it often remains agnostic about why certain structures are preferred or dispreferred in actual language use. This is where the cognitive framework provides its explanatory power. It illuminates the semantic and pragmatic motivations behind these syntactic possibilities, explaining how cognitive salience (i.e. the relative ease with which a concept comes to mind), $N_{possessor}$-$N_{possessee}$ dependency, and discourse accessibility constrain speakers' choices within the generative bounds. Specifically for this study, the generative model can inform us that a gapless relative clause like (2a) is syntactically well-formed, but it is the cognitive model that explains why a similar structure like (4b) fails. The latter lacks the necessary cognitive contiguity and low salience of $N_{possessee}$ to be construed as a defining property of $N_{possessor}$. Another typical example is that while the generative framework justifies when 的 is (non)mandatory through syntactic movement and lexical properties, the cognitive framework explains why 的 is more likely to be omitted in inalienable genitive constructions (e.g. kinship relation) due to strong conceptual relevance between $N_{possessor}$ and $N_{possessee}$. To date, coherent analyses of semantic-pragmatic constraints on Chinese genitive constructions under the cognitive framework remain under-documented.

What remains underexplored in existing accounts of 的 is its diagnostic value: because 的 marks the boundary between modifier and head, its (non)occurrence can serve as a probe into whether a given modifier-head relation is construed as conceptually dependent or independent at the moment of utterance. Under this view, 的 is not merely a syntactic head or an optional particle, but a morphosyntactic reflex of reference-point construal (Langacker, 1993). This reframing shifts the explanatory burden: instead of asking when 的 is required, I ask what the presence or absence of 的 reveals about the cognitive relation a speaker intends between $N_{possessor}$ and $N_{possessee}$. The upcoming analysis demonstrates that this diagnostic aligns with independent semantic-pragmatic constraints across predicate types, unifying phenomena previously treated separately.

[9] I treat head-raising as one viable generative "skeleton" for characterizing syntactic configurations of 的-marked genitive relative constructions. My semantic-pragmatic constraints for gradient acceptability do not necessarily hinge on the validity of head-raising itself. Were another syntactic derivation to replace it, the proposed cognitive-functional licensing conditions would still hold: generative syntax delimits the set of well-formed candidates, while cognitive-functional factors modulate their naturalness.

Crucial empirical contrasts motivating the present study (e.g. why (25a) is fully acceptable while (25b) is marginal, or why (29a) patterns differently from (29b)) remain unaccounted for by any existing framework. Both members of each pair are syntactically well-formed under standard generative analyses; the DP hypothesis and head-raising alike predict no contrast. Existing cognitive research has explored the semantics of the genitive relation in broad terms, yet no study to date has systematically investigated how the interaction between predicate type (adjectival versus verbal), noun individuation, and discourse accessibility yields gradient judgments across syntactically parallel structures. This is the research gap the present study addresses.

The research significance extends beyond Chinese-specific description and engages with a fundamental theoretical divide. Genitive constructions sit at a critical interface between syntax, semantics, and pragmatics, offering a crucial testing ground for questions of how grammatical structure interacts with cognitive and communicative principles. Regarding the two predominant schools of thought, I adopt a necessary hybrid perspective. While I accept the precise structural skeleton (such as the head-raising analysis for standard relatives and the clausal source of pre-的 modifiers), I move beyond the pure lexical-syntactic account by contributing a principled understanding of the cognitive salience and interpretive relevance of $N_{possessor}$-$N_{possessee}$ relation in human cognition. This principle integrates semantic tightness with pragmatic context. Consequently, this work is not merely an analysis of Chinese 的-constructions; it provides a replicable model for understanding how usage-based pressures selectively activate abstract syntactic templates. By examining why syntactically parallel Chinese forms differ in acceptability, this study contributes to the study of the syntax-semantics-pragmatics interface. It shows that the grammar licenses a structure while gradient acceptability is determined by non-syntactic factors such as reference-point accessibility, agentive interference, and discourse identifiability. Interface conditions shape usage-level preferences in a way that a purely syntactic account falls short.

One might ask why the present study does not adopt the constructional approach, which is implemented by Zhang and Wang (2025) as reviewed. Construction grammar treats form-meaning pairings as relatively fixed conventions stored in the linguistic system. It is well-suited for mapping the semantic range of a structure. However, the phenomenon here, where acceptability varies with minimal contextual shifts, can’t be reduced to whether the structure is listed as a construction. What

matters is whether the predicate-possessee combination can be construed as a diagnostically relevant property of $N_{possessor}$ within available cognitive frames, whether supplied by default world knowledge or by situational context (on-the-spot discourse situation). This is a problem of contextual licensing, not semantic taxonomy. Cognitive grammar, with its emphasis on reference-point construal (Langacker, 1993), captures this flexibility, whereas the focus of construction grammar on stored conventions does not offer a direct mechanism for context-driven acceptability shifts.

The present study contributes to the understanding of how the semantic relation between $N_{possessee}$ and $N_{possessor}$ and predicate attributes of relative clauses may affect the grammaticality of genitive relative constructions. I hereby discuss both adjective and verb genitive relative constructions by addressing two inquiries:

- What are the semantic-pragmatic constraints on adjectival-predicate and verbal-predicate genitive relative constructions?
- What are the constructional discrepancies between adjectival-predicate and verbal-predicate genitive relative constructions?

The preliminary hypothesis is that the felicity of genitive relative constructions is governed by the degree to which $N_{possessor}$ can function as an effective cognitive reference point for $N_{possessee}$. A reference point is a conceptually salient entity that provides mental access to a target entity. For genitive relative constructions, $N_{possessee}$ serves as the reference point, and $N_{possessor}$ as the target. Crucially, I predict systematic variation in acceptability among syntactically well-formed structures based upon semantic and pragmatic factors that are external to the syntactic module. A possessee that is highly individuated, volitional, or pragmatically independent from a possessor might yield lower acceptability, not due to a syntactic violation, but due to a failure in reference-point construal. The cognitive framework thus accounts for gradient judgments and pragmatic infelicity in a way that generative syntactic analysis, which delimits structural possibilities but remains neutral to usage-based preferences, fundamentally can't. Its predictive power lies in explaining not only what is syntactically well-formed, but also which structures are more readily accepted because they align with default cognitive patterns of categorization (conceptually natural) and which structures facilitate smoother information transfer by optimally balancing speaker encoding and hearer

decoding efforts (communicatively efficient).

Grammatical data discussed in the present study were collected from authentic Chinese newspapers and digital media. The sources include *People's Daily* (2016-2025) and *The Beijing News* (2018-2025). All constructed items were rated on a 7-point scale (1=completely unacceptable, 7=fully natural) by 20 native Mandarin speakers (aged 20-30, mean=25.4, recruited from universities in northern China. They reported Mandarin as their dominant language and had no history of speech or hearing disorders. A total of 33 critical stimuli were used. The stimulus set includes 12 planned minimal pair contrasts spanning 23 test items: 11 contrasts follow a standard two-item minimal pair format, while one contrast set involves three items ((41a), (41b), (41c)) contributing two separate comparisons. As 41(b) participates in two separate comparisons (see 41(a) versus 41(b) and 41(b) versus 41(c)), this shared item is counted only once in the tally of 23 unique comparison set sentences. Another 10 items were presented without minimal counterparts. In addition to 33 target stimuli, 15 filler items as adjunct-type gapless relative constructions were presented (similar to (8)), which fall outside the scope of the two genitive relative subtypes investigated in this study. This manipulation could prevent participants from detecting the experimental focus. All participants evaluated the full set of stimuli, which were presented in individually randomized order via Prolific.

For minimal pairs, paired t-tests were used to determine whether the predicted contrasts were statistically reliable ($p<0.05$); acceptable and unacceptable items within each pair were thus those forming significantly higher or lower groups. Across the 11 minimal pairs, the predicted acceptable items received significantly higher ratings (M=6.14, SD=0.63) than their unacceptable counterparts (M=3.18, SD=0.89). A paired t-test on the mean ratings collapsed across pairs confirmed a robust overall effect ($t(19)=9.86$, $p<0.001$, Cohen's $d=1.14$), and the contrast was significant for each individual pair (all $p<0.05$ after Bonferroni correction). Within minimal pairs, a subset of the lower-rated items exhibited comparatively greater between-rater dispersion and were accordingly marked as marginal ("?") rather than unacceptable ("*").

10 items without minimal counterparts were classified based upon rating distributions, including (31), (32), (35), (36), (40), (42), (43), (44), (45), and (46). Those showing a stable concentration of

low scores across raters were treated as unacceptable, whereas those with comparatively greater between-rater dispersion---operationalized as falling in the upper quartile of standard-deviation values across the stimulus set---were treated as marginal (marked as "?"). "?" indicates reduced acceptability in out-of-context presentation rather than ungrammaticality. The rating of such examples may substantially improve when discourse context provides the necessary cognitive framework for construing a reference-point relation between the head noun and the clausal subject. Among 10 items, 9 fall into the fully acceptable group, yielding a group-level mean rating of 5.81 (SD=0.68). One item, which is (32), is marginal, with a mean rating of 3.40 (SD=1.71).

5. Constraints On $N_{possessee}$+Adj+GEN+$N_{possessor}$

5.1 Inanimacy of $N_{possessee}$

When $N_{possessee}$ is animate, $N_{possessee}$+Adj+GEN+$N_{possessor}$ is difficult to be grammatical. (22a) and (23a) are not allowed to transform into (22b) and (23b) respectively. In the genitive relative relation, regardless of having a belonging relation (22a→22b) or a social relation (23a→23b), a living entity as a possessor can hardly appear in the subject position of a relative clause. Although 学生 'student' in (23b) is a monovalent noun that can provide an argument gap for another noun (Kou & Yuan, 2017), it is unable to form grammatical relative clauses like other monovalent nouns with whole-part or entity-attribute features.

Animacy alone, however, can't account for the contrast between (23b) and (24b), as 臣民 'subjects' in acceptable (24b) is likewise animate. The key lies in whether the predicate-possessee complex can function as a defining property of $N_{possessor}$. Subjects' loyalty forms an institutional relational feature of a king, making the clause an inherent characterization of the head noun. By contrast, 刻苦 'hard-working' denotes a student's volitional individual quality, which does not intrinsically define the coach.

Drawing on research on categorization (Cohen & Lefebvre; Du et al., 2023; Taylor, 2003), I propose that animacy imposes what can be termed agentive interference: animate entities are cognitively pre-categorized as potential agents, endowed with autonomous intentionality and

dynamic event potential. This pre-categorization conflicts with the attributive construal required by the adjectival-predicate genitive, which demands that $N_{possessee}$ should be mentally framed as a property carrier rather than an agent. (24b) is felicitous because the king-subject institutional bond suppresses such agentive construal. The underlying semantic constraint, concerning the conceptual dependency of $N_{possessee}$, is unpacked in section 5.2.

(22a) 老师 的 乌龟 很 害羞
lǎo shī de wū guī hěn hài xiū
teacher GEN turtle very shy
'The teacher's turtle is very shy.'

(22b) *乌龟 害羞 的 老师
wū guī hài xiū de lǎo shī
turtle shy GEN teacher
'the teacher with a shy turtle'

(23a) 教练 的 学生 很 刻苦
jiào liàn de xué shēng hěn kè kǔ
coach GEN student very hardworking
'The coach's student is very hardworking.'

(23b) *学生 很 刻苦 的 教练
xué shēng hěn kè kǔ de jiào liàn
student very hardworking GEN coach
'the coach with a hardworking student'

(24a) 国王 的 臣民 很 忠诚
guó wáng de chén mín hěn zhōng chéng
king GEN subject-PL very loyal
'The king's subjects are loyal.'

(24b) 臣民 很 忠诚 的 国王

chén mín hěn zhōng chéng de guó wáng
subject-PL very loyal GEN king
'the king whose subjects are loyal'

5.2 Dependency of $N_{possessee}$

In addition to animate $N_{possessee}$ that can hardly enter genitive relative constructions, some inanimate ones are not allowed to do so as well. Inanimate objects can be more readily associated with the stable property-ascribing function of the adjectival modifier. This is because inanimate objects often have more fixed and perceivable attributes that can be used to describe $N_{possessor}$. 屏幕 'screen' and 遥控器 'controller' are conceptually related to 电视 'television' in (25), so are 外观 'appearance' and 介绍 'introduction' to 艺术品 'artwork' in (26). As monovalent nouns, they must depend upon another noun. In both cases, (a) is more acceptable than (b) because $N_{possessor}$ and $N_{possessee}$ have greater conceptual closeness. At a more abstract level, I define dependency as not merely a spatial or functional association, but the degree to which $N_{possessee}$, mediated by the predicate, can serve as a defining attribute of $N_{possessor}$. For instance, 遥控器 'controller' and 电视 'television' have a functional relationship, but the attributes of a controller are less central to defining a television compared to a screen. This semantic hierarchy affects the acceptability of the construction.

(25a) 屏幕 精致 的 电视
píng mù jīng zhì de diàn shì
screen delicate GEN television
'the television with a delicate screen'

(25b) ? 遥控器 精致 的 电视
yáo kòng qì jīng zhì de diàn shì
controller delicate GEN television
'the television with a delicate controller'

(26a) 外观 有趣 的 艺术品
wài guān yǒu qù de yì shù pǐn
appearance interesting GEN artwork
'the artwork with an interesting appearance'

(26b) ？介绍 有趣 的 艺术品
jiè shào yǒu qù de yì shù pǐn
introduction interesting GEN artwork
‘the artwork with an interesting introduction’

What happens when $N_{possessee}$ is non-monovalent and bears a transferable relation to $N_{possessor}$? Both nominals in (27a) and (27b) instantiate non-monovalent $N_{possessee}$, yet they differ in relational type. (27a) yields a locational reading, whereas clothes in (27b) can encode a genitive possessive link with the athlete. Clean clothes constitute a contextually salient, relevant characteristic of an athlete, which improves the semantic coherence of this construction. This pair thus illustrates how the nature of the dependency between $N_{possessor}$ and $N_{possessee}$ distinguishes locational from genitive interpretations.

(27a) ？桌子 干净 的 运动员
zhuō zi gān jìng de yùn dòng yuán
table clean GEN athlete
‘the athlete with a clean table’

(27b) 衣服 干净 的 运动员
yī fú gān jìng de yùn dòng yuán
clothes clean GEN athlete
‘the athlete with clean clothes’

The requirement for a close semantic fit finds parallels in cross-linguistic data with grammaticalized noun classes. In the Bantu language Luganda, the genitive concord morpheme agrees in noun-class with $N_{possessee}$, formally indexing the possessed nominal (28). While this morphological marking encodes possessive linkage, it does not grammatically enforce conceptual dependency between $N_{possessor}$ and $N_{possessee}$.

(28) e-ki-tabo ky-a Muganda
AUG-CL7-book 7.POSS-of 1.Muganda
‘the book of a Muganda person’

5.3 Abstraction of $N_{possessee}$

The naturalness of $N_{possessee}$ is also subject to other semantic factors, such as noun abstraction. (29a) and (30a) have better grammaticality than their counterparts. 家具 'furniture' and 植物 'plant' are the superordinate category, while 衣柜 'closet' and 柏树 'cypress' are the basic category. Nouns describing the superordinate category are more abstract than those featuring the basic category. Joosten (2010) argues that speakers employ nouns with the superordinate category in a non-specific way by intentionally reducing the prominence of individual members and focusing on abstract similarities between them. This idea suggests that speakers are more concerned with attributes than entities when using superordinate category nouns. Things that appear as the basic category usually have a higher degree of individuality, so nouns with clear referents and a high degree of individuality are less likely to enter genitive relative constructions, whereas nouns with descriptive, macroscopic attributes are allowed by this structure. Descriptive and semantically general nouns among monovalent ones are highly abstract; grammatical genitive relative constructions formed between such nouns and clausal heads may be accounted for by the level of abstraction (see (29a) and (29b)).

The cognitive underpinning of this abstraction effect lies in conceptual zoom-out: superordinate nouns (e.g. 家具 'furniture') activate a schematic mental representation (Wang & Zhao, 2024) that foregrounds category-level attributes while backgrounding individual exemplar features. This schematic mode of reference aligns with the attributive function of adjectival-predicate genitives, where $N_{possessee}$ serves to characterize $N_{possessor}$ via generic properties rather than specific instances. Basic-level nouns (e.g. 衣柜 'closet'), by contrast, trigger exemplar-specific mental simulation, evoking detailed perceptual and functional imagery that resists integration into a purely attributive frame. The gradient acceptability thus reflects a match or mismatch between the level of conceptual specificity recruited by the noun and the attributive construal demanded by the construction.

(29a) 家具 整齐 的 房间
jiā jù zhěng qí de fáng jiān
furniture neat GEN room
'the room with neat furniture'

(29b) ？衣柜　整齐[10]　的　房间
yī guì　zhěng qí　de　fáng jiān
closet　neat　GEN　room
‘the room with a neat closet’

(30a) 植物　茂盛　的　花园
zhí wù　mào shèng　de　huā yuán
plant　luxuriant　GEN　garden
‘the garden with luxuriant plants’

(30b) ？柏树　茂盛　的　花园
bǎi shù　mào shèng　de　huā yuán
cypress　luxuriant　GEN　garden
‘the garden with luxuriant cypresses’

Moreover, some nouns with individual characteristics may sometimes enter genitive relative constructions with adjectives such as 多 ‘many’ and 丰富 ‘abundant’. The predicate 多 or 丰富 directly endows its subject with a generic referential feature (Liu, 2002). Generic reference usually focuses more on the connotation of the referent than on its denotation by reducing individuality. In this way, its adjacent noun still tends to denote a more abstract attribute (Liu, 2002; Wang, 2005). Hence, 鸟儿 ‘bird’ and 饭菜 ‘dish’ as the basic category are also allowed in genitive relative constructions (see (31) and (32)).

(31) 鸟儿　很多　的　天空
niǎo ér　hěn duō　de　tiān kōng
bird　many　GEN　sky
‘the sky with many birds’

(32) ？饭菜　丰富　的　大厅
fàn cài　fēng fù　de　dà tīng

[10] Replacing 整齐 ‘neat’ in (29b) with 超大 ‘super big’ may improve acceptability. Hence, the constraint is not merely about noun abstraction (basic versus superordinate level), but about whether the predicate-$N_{possessee}$ combination can be construed as a defining characteristic of $N_{possessor}$. “Super big closet” can readily characterize a room, whereas “neat closet” typically can’t. Thus, interpretive relevance (as previously discussed) can override default abstraction preferences.

dish abundant GEN lobby

'the lobby with abundant dishes'

5.4 Cognitive framework between $N_{possessor}$ and $N_{possessee}$

For $N_{possessee}$+Adj+GEN+$N_{possessor}$, $N_{possessee}$ has complex semantic constraints that can help to construct a link between a relative clause and $N_{possessor}$. These constraints contain three domains: inanimacy, dependency, and abstraction. Among them, whereas inanimacy and abstraction point more to the semantic characteristics of $N_{possessee}$ itself, dependency operationally defines contextually modulated property relevance between $N_{possessor}$ and $N_{possessee}$ rather than the alienable/non-alienable or divisible/indivisible relation that has been the focus of other studies of genitive constructions (e.g. Carnesale, 2022; Cohen, 2019; Öztürk & Taylan, 2016). The effect of dependency on the grammaticality of genitive constructions reflects the value of dependent semantics in typological research. Fox (1981) found through cross-linguistic research that the essence of the whole-part relation can be attributed to the contiguous relation, not solely to the inalienable, indivisible relation. Syntactic structures with whole-part semantic constraints have traditionally been considered to accommodate alienable but contiguous $N_{possessor}$-$N_{possessee}$. (33) shows how the syntactic raising of $N_{possessor}$ in Spanish was once thought to occur only in whole-part relations (Fox, 1981). In this case, *le* 'her' is raised at the front.

(33) Le secó las lágrimas

3SG.DAT dry.3SG.PST DEF.PL tear.PL

'He dried her tears.'

(34) Le veo las enaguas

3SG.DAT see.1SG.PRES DEF.PL petticoat.PL

'I saw her petticoat.'

However, possessor-raising constructions may also manifest the pure contiguous relation (Fox, 1981). (34) shows *le* 'she' and *enaguas* 'petticoat' do not have a whole-part relation, but they are still allowed to enter this possessor-raising construction. Hence, even if $N_{possessee}$ is lexically self-contained and non-monovalent, the proximity or contiguity between $N_{possessor}$ and $N_{possessee}$ can easily be perceived as having a "whole-part" relation through grammaticalization. Cognitively, the

dependent relation has a wider scope of event coverage and more explanatory power than the whole-part relation. Parallel to monovalent nouns, the dependent relation reveals a similar level of genitive affinity and mutual realization of cognitive activation between $N_{possessor}$ and $N_{possessee}$.

No matter the semantic relation of $N_{possessor}$-$N_{possessee}$ represented by dependency or the semantic features characterized by inanimacy and abstraction, they both embody the degree of individuality of nouns, which in turn can reflect the extent of cognitive salience of relevant events. Events that are salient at the cognitive level tend to be more semantically individual and occupy the basic category. Syntactically, marked forms are evident. Salience has corresponding manifestations at the semantic and syntactic levels (Gu, 2001). The lexical form, deictic content, referential form, and quantitative form of a noun all affect its salience (Givón, 1976; Hopper & Thompson, 1980; Deane, 1987; Korzen, 2015) (**Table 1**).

Table 1. Features that affect the salience of nouns

	salience	non-salience
deictic form	pronoun (1st & 2nd person>3rd person)	noun
deictic content	animacy (human>animal)	inanimacy (concreteness>abstraction)
quantitative form	countability (single>plural)	uncountability>collective set
referential form	definiteness	generic>indefinite>no reference

$N_{posssessee}$ that can act as the subject in a genitive relative construction is characterized by low salience, mainly in terms of its inanimate and abstract features. Nouns with low salience usually do not refer to definite entities but are abstract concepts describing event properties referred to by their clausal head, thus creating a dependency relation and forming a qualified genitive relative construction. In $N_{possessee}$+Adj+GEN+$N_{possessor}$, relative clauses are usually used to portray certain solid attributes of their noun heads, which requires that $N_{possessee}$ must best represent the characteristics of $N_{possessor}$. Shi (2002) distinguishes the semantic structure of nouns into referential and descriptive features, arguing that the latter is the part of noun semantics that represents stable properties. Moreover, referential features are mainly found in concrete nouns, while abstract nouns are characterized by descriptive features. Thus, in genitive relative constructions that undertake the descriptive function, inanimate and abstract $N_{possessee}$ is more likely to be permitted.

The semantic features affecting the eligibility of genitive relative constructions can be further categorized as dependency and low salience. Two conclusions can be drawn. First, the conceptual pairs of low salience and attachment are intertwined, and together they have a constraining effect upon genitive relative constructions. When the noun is highly dependent (typically a whole-part monovalent noun), the abstraction of $N_{possessee}$ is not required, which can be a concrete entity (see (25a)). When the concept denoting $N_{possessee}$ is relatively independent, divisible, or alienable, $N_{possessee}$ needs to elevate its abstraction, allowing the whole construction to be more grammatical (see (30b)→(30a)).

Second, low salience and dependency are two different pairs of semantic concepts that describe different relationships between $N_{possessor}$ and $N_{possessee}$. The former shows the connotative dependence of $N_{possessee}$ on $N_{possessor}$, as the speaker needs to understand and perceive the concepts expressed by $N_{possessee}$ through other entities. The latter establishes the dependent relation between $N_{possessor}$ and $N_{possessee}$ through interpretive relevance. The commonality is that both show $N_{possessor}$-$N_{possessee}$ affiliation; $N_{possessee}$ with low salience and dependency lacks independent recognition, and so its $N_{possessor}$ needs to be co-activated with a shared cognitive framework for better perception.

A noun with low salience or dependency does not contain an obligatory dependence on another noun component as a monovalent noun does (Yuan, 1994). Such nouns are lexically self-contained (e.g. 时间 'time', 衣服 'clothing', 家具 'furniture'). On the other hand, they are (optionally) dependent upon their $N_{possessor}$ on a semantic basis. As the pair of $N_{possessor}$-$N_{possessee}$ frequently co-occurs in the same cognitive frame, $N_{possessee}$ is used to characterize stable features of its $N_{possessor}$. Dependency and low salience as semantic conditions constraining $N_{possessee}$+Adj+GEN+$N_{possessor}$ are a mixture of lexical meanings (i.e. whole-part nouns, abstract nouns) and contextual meanings (temporarily created dependent meaning). I propose that this mixture reflects two distinct cognitive operations: semantic dependency arises from lexical-conceptual structure stored in long-term memory (e.g. the relational schema for screen evokes television as its default reference point), whereas contextual dependency emerges from online inferencing in working memory, where a predicate-$N_{possessee}$ combination is temporarily construed as characteristic of $N_{possessor}$ within a given discourse frame. Grammaticality judgments become gradient precisely when these two operations conflict, namely when lexical semantics resists the imposed contextual relation and thereby requires

additional pragmatic support (e.g. focal adverbs, determiners) to resolve the tension.

In sum, **Table 2** below summarizes the semantic constraints on adjectival predicates embedded in genitive relative constructions.

Table 2. The semantic constraints on $N_{possessee}$+Adj+GEN+$N_{possessor}$.

constraint type	requirement
inanimacy	$N_{possessee}$→typically inanimate
dependency	contextually modulated property relevance
abstraction	preference for superordinate or generic nouns over specific referents
cognitive salience	low referential salience

The three constraints do not carry equal weight, nor do they operate independently. Rather, they form an implicational hierarchy: dependency is the primary licensing condition, while inanimacy and abstraction function as facilitative rather than absolute requirements. When the dependency relation between $N_{possessor}$ and $N_{possessee}$ is strongly entrenched---as with monovalent part-whole nouns, kinship terms, or socially institutionalized role pairs---the construction is licensed even if the possessee is animate. In such cases, the cognitive salience of the dependency relation overrides the default preference for an inanimate, abstract possessee. Conversely, when the dependency relation is weak or absent---as with alienable, contextually unconnected pairs---the construction places heavier demands on the inherent semantic properties of $N_{possessee}$, requiring low individuation and high abstractness to compensate for the lack of a pre-established cognitive frame. The hierarchy thus predicts that acceptability is highest when all three constraints align, remains high when dependency overrides inanimacy or concreteness, and degrades when neither dependency nor abstraction is available to support the reference-point construal (Langacker, 1993).

6. Constraints On $N_{possessee}$+VP+GEN+$N_{possessor}$

While adjectival-predicate genitive relative constructions are heavily constrained by semantic factors such as dependency and salience, verbal-predicate constructions operate under a different set of conditions, primarily driven by pragmatic and discourse considerations. In verbal-predicate

genitive relative constructions, the semantics is centered around events. $N_{possessee}$ does not need to have the same kind of static, inherent relationship with $N_{possessor}$ as in the adjectival-predicate structure. Instead, it is about the event encoded by the verb and its relation to $N_{possessor}$.

(35) 数据 没 上传 的 员工
shù jù méi shàng chuán de yuán gōng
data NEG update GEN employee
'the employee whose data are not uploaded'

(36) 奖金 被 没收 的 球员
jiǎng jīn bèi mò shōu de qiú yuán
bonus PASS confiscate GEN player
'the player whose bonus is confiscated'

In (35) and (36), $N_{possessee}$ (数据 'data', 奖金 'bonus') appears in verbal-predicate genitive relative constructions. While such nouns can occasionally appear in adjectival-predicate constructions (e.g. 奖金很多的员工 'the employee with much bonus', 数据丰富的研究 'the study with rich data'), their acceptability is highly constrained. They are licensed primarily when the adjective denotes a scalar or quantitative property (e.g. 多 'many', 丰富 'abundant', 少 'few') that can be construed as a measurable attribute of the possessor as a whole (e.g. an employee's bonus level, the data richness of a study). This restricted adjectival compatibility contrasts with the wider range of descriptive adjectives (e.g. 好看 'good-looking', 精致 'delicate') freely allowed with inanimate, dependent possessees like 屏幕 'screen'. Therefore, the core distinction remains: verbal predicates are inherently eventive, while adjectival predicates are attributive; but the semantic class of the adjective (e.g. quantitative versus descriptive) interacts with the noun properties, affecting acceptability within the adjectival-genitive type. This nuance further supports the centrality of interpretive fit between predicate and possessee-possessor relation.

It is noted that if $N_{possessee}$ is animate like (37), some verbal constructions are still ungrammatical. By comparing (35), (36), and (37), we find that low salience does not guarantee grammaticality. Adjectival and verbal genitive relative constructions differ in semantic conditions and discourse functions. In (35), the focus is on the event of the data not being uploaded and how it is associated

with the employee. The semantic role of $N_{possessee}$ here is more about being the theme in the event, rather than representing an inherent property of $N_{possessor}$. When $N_{possessee}$ is animate, as in (37a), the ungrammaticality might arise because the autonomous behavior of animate $N_{possessee}$ (the bird's screaming) is difficult to integrate into an event that definitionally characterizes the possessor. It could be that the event of the bird screaming is too independent and not easily construed as a defining event for the servant in the same way as data-related events for an employee. The acceptability of (37b) is marginally higher than that of (37a), despite both subjects being animate. The event "was stolen" is an externally caused event that directly affects the interests of $N_{possessor}$. The residual marginality of (37b), however, indicates that eventive type alone does not fully license the construction. What is additionally required is a conventional cognitive frame linking an event to the identity of $N_{possessor}$, a frame present in the employee-data relation in (35) but absent in the servant-bird relation (even when the eventive type is controlled).

(37a) *鸟 尖叫 的 仆人
niǎo jiān jiào de pú rén
bird scream GEN servant
'the servant whose bird screams'

(37b) *鸟 被 偷 走 的 仆人
niǎo bèi tōu zǒu de pú rén
bird PASS steal away GEN servant
'the servant whose bird was stolen'

The informational function carried by relative clauses in discourse has been deeply explored by different scholars from different perspectives. From the perspective of informational status, relative clauses can be divided into informative relative clauses and non-informative relative clauses (Bernardo 1979). An informative relative clause can provide new information about a referring noun and describe some properties of a definite head noun. Non-informative relative clauses are divided into identificatory relative clauses and specificatory relative clauses (Fox & Thompson, 1980). Neither adds information about the noun. Within this non-informative group, the events described by relative clauses are treated as known to the listener. The difference is that an identificatory relative clause refers back to a definite noun, and a specificatory relative clause

defines an indefinite noun through a known event. See **Table 3** for a summary.

Table 3. Classification of relative clauses

relative clause	informative relative clause	
	non-informative relative clause[11]	identificatory relative clause
		specificatory relative clause

Fang and Song (2004) distinguish relative clauses into naming, introducing, and tracing. Among them, tracing clauses correspond functionally to identificatory clauses (Bernardo, 1979), and the function of naming clauses and specificatory clauses is basically the same. On the other hand, in an introducing relative clause, both clausal relativeness and the noun are new information; the head is definite, which is not in the framework proposed by Bernardo (1979). Therefore, I operationally label the information types in genitive relative constructions: information adding, tracing (identificatory clauses), naming (specificatory clauses), and introducing. Regarding the coding criteria, a relative clause was coded as information adding if it supplied new descriptive content about an already identifiable head noun, enriching a hearer's knowledge without being necessary for referent identification (e.g. see (25)). A clause was coded as tracing if it served to re-identify a previously mentioned referent by evoking a known event uniquely associated with that referent (e.g. see (36)). A clause was coded as naming if it provided a specificatory event that narrows down the denotation of an indefinite or generic head noun (e.g. see (45)). A clause was coded as introducing if both the relative clause and the head noun conveyed new information, with the head noun being definite. It is a category posited by Fang and Song (2004) but unattested in the present corpus sample.

To quantify the distribution of discourse functions between adjectival and verbal-predicate genitive relative constructions (cf. **Table 4**), a corpus-based analysis was conducted by using The Center For Chinese Linguistics PKU (CCL) corpus. The search was conducted by using the part-of-speech tagged version of CCL. For adjectival-predicate genitive relative constructions, the query pattern "[n] [a] 的 [n]" was employed, followed by manual filtering to exclude non-genitive relative uses (e.g. predicative or comparative structures[12]) and to verify that the first noun functioned as the

[11] All adjectival-predicate genitive-relative examples in this study function as identificatory relative clauses. Specificatory readings are attested in ordinary Mandarin restrictive relative clauses, which lie outside the target construction examined here.

[12] In predicative structures like 他买的东西很贵的品牌 'the brand of the things he bought being expensive', the substring 东西很

subject of the adjective. For verbal-predicate genitive relative constructions, the query "[n] [v] 的 [n]" was used, with similar manual verification to ensure a genuine subject-predicate relation within the clause and to exclude cases where a verbal element is part of a larger auxiliary or serial verb construction[13]. The raw search returned 427 hits for the adjectival-predicate pattern and 519 hits for the verbal-predicate pattern. After removing ambiguous, fragmentary, and non-target cases, 312 eligible adjectival tokens and 374 eligible verbal tokens remained in the filtered pool. From these eligible pools, 100 tokens per type were randomly sampled from their respective filtered pool. I coded each instance for its primary discourse function (information adding, tracing, naming, introducing), and a 20% subset was independently coded by a second trained annotator, achieving an inter-annotator agreement (Cohen's κ) of 0.79.

Table 4. Discourse functions: $N_{possessee}$+Adj+GEN+$N_{possessor}$ and $N_{possessee}$+VP+GEN+$N_{possessor}$

genitive relative type	information adding	tracing	naming	introducing	total
adjectival	83 (83%)	3 (3%)	14 (14%)	0 (0%)	100
verbal	0 (0%)	81 (81%)	19 (19%)	0 (0%)	100

As we can observe, the main function of $N_{possessee}$+Adj+GEN+$N_{possessor}$ is to supplement information, while $N_{possessee}$+VP+GEN+$N_{possessor}$ is mostly used to describe known events. The latter is often not a substitute for the former regarding information supplementation.

(38a) ？书包 丢了 的 学生
shū bāo diū le de xué shēng
schoolbag lose-PFV GEN student
'the student whose schoolbag is lost'

(38b) 书包 沉重 的 学生
shū bāo chén zhòng de xué shēng
schoolbag heavy GEN student
'the student with a heavy schoolbag'

贵的 'things that are expensive' functions as a predicate to the head noun 品牌 'brand', not as its genitive modifier. Similarly, comparative constructions such as 比他高的男孩 'the boy taller than him' were excluded because the adjective phrase 比他高的 'taller than him' is headed by a comparative operator, not a genitive constituent.

[13] A typical example when a verbal element is part of a larger auxiliary is 能来的人 'people who can come'. The verbal unit 能去 'can come' consists of an auxiliary verb+main verb, forming a single modal predicate. Likewise, in serial verb constructions like 去买菜的老人 'the old person who goes to buy groceries', the sequence 去买菜 'goes to buy' constitutes a multi-verb event. They do not represent the clause structure under investigation.

$N_{possessee}$+VP+GEN+$N_{possessor}$ is usually used to track and confine $N_{possessor}$. When using this construction, $N_{possessor}$ needs to be identifiable in the context. As a minimal pair, (38b) is more acceptable than (38a). In (38b), the phrase 沉重 'heavy' adds a descriptive layer to the schoolbag, which can help the listener better visualize or identify the student in the context. In contrast, the verbal-predicate constructional type is rarely used in situations where $N_{possessor}$ first appears like (38a). The verbal predicate 丢了 'is lost' denotes a transient, episodic event. As such, it is difficult to construe as a stable, inherent property of the possessor 学生 'student'. This contrast underscores a key constraint on verbal-predicate genitive relative constructions: they are dispreferred when the event denoted by the verb is not highly accessible in discourse.

I propose a deeper cognitive asymmetry in event schematization (Langacker, 2013). Adjectival predicates invoke a state schema, where the predicate-possessee complex is mentally represented as a temporally stable property anchored to the possessor. Verbal predicates, by contrast, invoke an episodic schema, representing the predicate-possessee complex as a temporally bounded occurrence with its own internal structure. The cognitive cost of integrating an episodic schema into a modifying function arises from schema competition: episodic representations privilege temporal specificity and causal structure, which compete with the attributive requirement that the modifier serve as a stable identification cue. Verbal-predicate genitive relative constructions are licensed only when contextual cues (e.g. discourse-old status, focal adverbs) override this default incompatibility by foregrounding the identificatory function of the event.

Verbal-predicate genitive relative constructions are not as commonly used in existential sentences without sufficient context compared with adjectival-predicate genitive relative constructions, like (39). (39b) is more acceptable than (39a). The key difference lies in the framing property of the predicate: the verb 破碎 'fracture' encodes a change-of-state event (the window became broken), which is episodic and implies a prior event. In an existential sentence introducing a new referent, such an eventive modifier is less natural because it shifts focus from the property of the possessor (the room) to a past occurrence affecting the possessee. In contrast, the adjective 很大 'very big' describes a stable, enduring property of the window, which readily serves to characterize the room in a modifying function. Alternatively, an existential sentence turns out to be acceptable if $N_{possessor}$

or $N_{possessee}$ is more accessible through a set pragmatic situation, like (40).

(39a) ？有 一个 窗户 破碎 的 房子
yǒu yí gè chuāng hu pò le de fángjiān
have one-CL window fracture GEN house
'There is a room with a broken window.'

(39b) 有 一个 窗户 很 大 的 房子
yǒu yí gè chuāng hu hěn dà de fángjiān
have one-CL window very big GEN house
'There is a room with a very big window.'

(40) 这 是 我 家, 有 一个 没 关 门 的 房间
zhè shì wǒ jiā yǒu yí gè méi guān mén de fáng jiān
this COP 1SG home have one-CL NEG close door GEN room
'This is my home, where there is a room with an open door.'

Appositive is a typical syntactic position for increasing the occurrence of informative relative clauses. The adjectival-predicate genitive relative construction like (41a) is rigorously grammatical. The relative unnaturalness of (41b) exemplifies a key constraint on verbal-predicate genitive relative constructions in appositive positions. Unlike adjectival predicates that denote stable properties, verbal predicates introduce eventive content whose felicity heavily depends on discourse coherence. In (41b), the modifying event "failed the exam" lacks a plausible situational connection to the main clause "left the toilet", creating a pragmatic mismatch that hinders naturalness. In (41c), once such contextual relevance is established by aligning the modifier with the main clause setting (the test room) and the protagonist's resulting state (in low spirits), the verbal-predicate genitive relative construction becomes fully acceptable. This contrast underscores that the licensing of verbal-predicate genitive relative constructions, especially in appositives, is not entirely syntactically prohibited but is highly sensitive to the interpretive relevance of the modifying event to the broader discourse context. It reinforces the core principle that verbal predicates, being eventive, require a greater degree of contextual integration to function felicitously as modifiers compared to property-denoting adjectives.

(41a) 杰克, 这个 举止 得体 的 少年, 离开了 卫生间
jié kè zhè gè jǔ zhǐ dé tǐ de shào nián lí kāi le wèi shēng jiān
Jack this behavior appropriate GEN teenager leave-PFV toilet
'Jake, the well-mannered teenager, left the toilet.'

(41b) ? 杰克, 这个 考试 失败了 的 少年, 离开了 卫生间
jié kè zhè gè kǎo shì shī bài le de shào nián lí kāi le wèi shēng jiān
Jack this exam fail-PFV GEN teenager leave-PFV toilet
'Jake, the teenager who failed the exam, left the toilet.'

(41c) 杰克, 这个 考试 失败了 的 少年, 离开了 考场
jié kè zhè gè kǎo shì shī bài le de shào nián lí kāi le kǎo chǎng
Jack this exam fail-PFV GEN teenager leave-PFV exam room
'Jake, the teenager who failed the exam, left the exam room.'

Two genitive relative types differ not only in syntactic representation (predicate type) and semantics (dependency or low salience of $N_{possessee}$), but also in discourse behaviour. Adjectival-predicate genitive relatives typically encode head-noun properties and serve both descriptive and differentiating functions, supporting a richer array of informational uses. By contrast, verbal-predicate genitive relatives draw on contextually salient events for their clausal content and perform a trace-identifying function. They permit greater flexibility in possessive semantic relations between $N_{possessor}$ and $N_{possessee}$, yet verbal predicates impose additional argument-structural constraints and substantial pragmatic requirements.

To justify the rationale of functional disparity, I hypothesize that in nature, adjectival-predicate genitive relative constructions are designed to support the presentation of stable properties and express the possession of an attribute by the head noun. They are rich in information because they can add descriptive details about the head noun, which can be new information in a discourse. This idea is in line with the given-new information principle, where new information is presented in a way that builds on the given context (see (38)). Speakers are thus allowed to evoke a mental image of the head noun with specific attributes, facilitating identification and differentiation. In verbal-predicate genitive relative constructions, the verb brings in its own set of arguments, and the relationship between $N_{possessor}$ and $N_{possessee}$ is mediated through an event denoted by the verb. It

seems that $N_{possessor}$ and $N_{possessee}$ have more flexible relationships with fewer constraints, but their binding must be central to a fixed event to help the listener grasp the discourse focus. This idea reflects the cognitive concept of event-based categorization (Ünal et al., 2021). We use events as a way to mentally group and identify entities. Since events are often more context-dependent than properties, verbal-predicate genitive relative constructions are more closely tied to specific contexts for their interpretation.

In sum, **Table 5** below compares the functional preference between adjectival- and verbal-predicate genitive relative constructions.

Table 5. The functional preference between $N_{possessee}$+Adj+GEN+$N_{possessor}$ and $N_{possessee}$+VP+GEN+$N_{possessor}$.

constructional type	primary discourse function
$N_{possessee}$+Adj+GEN+$N_{possessor}$	information supplementation
$N_{possessee}$+VP+GEN+$N_{possessor}$	information tracing/identification

7. Pragmatic Functions Operating On Genitive Relative Constructions

Wan (2012) and Zhang (2014, 2018) argue that Chinese noun-modifier structures pragmatically represent a reference-target relation between the modifier and the head. This viewpoint may explain many relative clauses in Chinese that can't be justified by purely syntactic relations, including genitive relative constructions.

(42) 车 开 得 最 快 的 司机
chē kāi de zuì kuài de sī jī
vehicle drive DEG most fast GEN driver
'the driver who drives the fastest'

(43) 挑战 最 大 的 手术
tiǎo zhàn zuì dà de shǒu shù
challenge SUP big GEN operation
'the most challenging operation'

Cognitively, a reference needs to be sufficiently salient to be narrowed down, which helps to search for a target entity. Salience comes from the high accessibility of a reference system. Accessibility refers to how easy or difficult a listener can search for and find the target event in their mind or surroundings after hearing a reference, as predicted (Winters & Nathan, 2020). An event may be highly atypical in terms of world knowledge yet highly accessible in the relevant sense, provided it carries sufficient diagnostic value to single out the referent. Thus, at the discourse level, the formation of relative structures (and other genitive relative structures) is limited by attributive accessibility. A modifier structure can be established when its attribute has a high degree of accessibility in the discourse and can be used as a reference to locate the head. For (40), the event described by the construction is more accessible in context. For (42) and (43), the adverb 最 'most', marking event focus, strengthens the identificatory force of this construction. Both the high accessibility of an event and the inclusion of an event focus may increase the identificatory degree of a relative clause, making the attribute better realize its identificatory function. On the other hand, $N_{possessee}$+VP+GEN+$N_{possessor}$ as a static description usually appears as background information; its attribute is less accessible in the context, making the construction more subject to pragmatic constraints. Hence, although both of them are genitive relative constructions, the eligible conditions of $N_{possessee}$+Adj+GEN+$N_{possessor}$ and $N_{possessee}$+VP+GEN+$N_{possessor}$ are subject to different linguistic levels. The former can be attributed to dependency or low salience of $N_{possessee}$ at the semantic level, while the latter is mainly affected by accessibility at the pragmatic level.

A further argument is that the semantic constraints of dependency and low salience can be generalized to accessibility as well. Wan (2012) proposes that the more an entity intrinsically characterizes another entity, the more likely it is to be used as a reference (see (27b) in the present study). This debate reflects the lexical-semantic relation between $N_{possessor}$ and $N_{possessee}$: when they have an implicit dependent relation, it is more likely that $N_{possessee}$ functions as a reference to describe $N_{possessor}$. Thus, higher accessibility of $N_{possessee}$ in this case may contribute to a more acceptable genitive relative construction. Under this account, the essence of the semantic constraints of $N_{possessee}$ is also the degree of accessibility.

Examples carrying the pragmatic function mostly need to depend upon appropriate contextual

situations. In (42) and (43), the clausal subjects have been activated in the context, especially through the addition of 最 'most', transferring the event focus to the content of their relative clauses and strengthening the differentiable function. The primary pragmatic function of a relative clause, especially in speech, is to narrow down the identification of the head noun. Once a relative clause needs to undertake a differentiable function, the construction may break through various syntactic and semantic constraints in relativization to form an eligible structure.

(44), (45), and (46) use unmarked relativization to form relative structures with oblique case. On a purely semantic basis, these sentences may violate noun accessibility and conventional relativization in Chinese (Comrie & Keenan, 1979; Kou & Yuan, 2019). All the sentences have a distinctive colloquial style with the adverbs 刚才 'just now' and 已经 'already' to suggest specific events. Moreover, a construction may insert a referential quantifier, like 那个 'that' in (46), constituting "clausal relative externalization", as this constituent is placed after the genitive marker (Tang, 2007).

(44) 我们 刚才 吃 饭 的 桌子
wǒ men gāng cái chī fàn de zhuō zi
1PL just eat meal GEN table
'the table where we've just had a meal'

(45) 学校 已经 发了 毕业证 的 学生
xué xiào yǐ jīng fā le bì yè zhèng de xué shēng
school already issue-PST testamur GEN student
'students who have been issued a testamur from the school'

(46) 捷克 输 20 分 的 那个 明星 队伍
jié kè shū 20 fēn de nà gè míng xīng duì wu
Czech lose 20 points GEN that star team
'the star team that defeated the Czech team by 20 points'

Although such instances may seemingly be an exception to previously discussed semantic constraints of genitive relative constructions, these relative structures serve the identificatory and

specificatory functions in speech. To facilitate a listener's quick recognition of what a speaker is referring to, a speaker may resort to reference events to quickly activate that listener's memory. Hence, this type of genitive relative construction typically draws on a focal constituent, encoding temporal-spatial attributes of a specific event. Such adverbs or determiners support the target genitive-relative reading. From (42) to (46), if we remove the focal information brought by 最 'most', the event-specific information brought by 刚才 'just' and 已经 'already', as well as the determinative, specificatory marker 那个 'that', the identificatory genitive-relative reading becomes far less salient, and competing non-genitive interpretations gain priority.

Differentiability has always been regarded as the most critical pragmatic function of relative clauses, especially in spoken corpora, in which substantial relative clauses function to trace and discern their head nouns (Tao, 2002). Lu (2003) argues that both descriptiveness and differentiability belong to adjunct in a broad sense. Descriptiveness modifies a core constituent connotatively, whereas differentiability and referentiality emphasize denotation. They represent two different levels: the former embodies a semantic function, and the latter reflects a pragmatic function.

Lu (2003) suggests that descriptiveness is the basic function while differentiability is the derived function. As a relative clause with a descriptive function at the semantic level, it usually has stronger syntactic-semantic constraints, such as an adjectival-predicate genitive relative construction in which the semantic constraints of $N_{possessee}$ are stricter. Among relative clauses, genitive relative constructions are descriptive at the semantic level and can also take on a differentiable function at the pragmatic level. Particularly at the pragmatic level, when a speaker needs to quickly refer to a specific entity, they may temporarily break through the syntactic-semantic constraints for the sake of communication by combining events (reference) and their participants (target) that are loosely related. This type of "temporary" relative structure is characterized by differentiability at the pragmatic level but hardly has descriptiveness at the semantic level. For genitive relative constructions, this limitation restricts them from taking on the function of information adding.

Temporary breakthrough can be understood as a reference-point shift driven by communicative urgency. Under normal conditions, reference-point selection is guided by cognitive optimality

(Lieder & Griffiths, 2020): speakers choose $N_{possessee}$ that are conceptually dependent and routinized. However, when rapid identification is required, the speaker may override these default preferences by constructing a novel reference-point relation online, leveraging shared context to ensure recoverability. Such constructions, illustrated from (42) to (46), trade off long-term routinization for short-term communicative efficiency, explaining why they are pragmatically licensed but semantically marked.

In sum, **Table 6** below outlines the pragmatic licensing conditions for both adjectival- and verbal-predicate genitive relative constructions.

Table 6. Pragmatic licensing conditions for genitive relative constructions.

constructional type	primary discourse function	licensing condition
$N_{possessee}$+Adj+GEN+$N_{possessor}$	information supplementation	high inherent accessibility of the modifier (via dependency/low salience)
$N_{possessee}$+VP+GEN+$N_{possessor}$	information tracing/identification	high contextual accessibility of an event

8. Conclusion

The present study originally contributes to our understanding of semantic-pragmatic constraints of Chinese genitive relative constructions by referring to adjectival-predicate and verbal-predicate structures. I hereby draw two conclusions. First, starting from $N_{possessee}$+Adj+GEN+$N_{possessor}$, I argue that this type of relative clause requires contextually modulated property relevance between $N_{possessor}$ and $N_{possessee}$, as well as inanimacy and abstractness of $N_{possessee}$. The latter two features can be further generalized as low salience. Regardless of dependency or low salience of a head noun, it serves to strengthen the dependent relation of $N_{possessor}$-$N_{possessee}$ pairing that can be activated in the same cognitive framework. In this way, $N_{possessee}$+Adj+GEN+$N_{possessor}$ can be natural.

Second, regarding $N_{possessee}$+VP+GEN+$N_{possessor}$, $N_{possessee}$ can be occupied by a noun carrying the basic category without semantic constraints. However, this constructional type is limited in its pragmatic functions and can't take on the function of information supplementation. I also analyze some constructions that may break through this constraint and find that those with pragmatic

differentiability are less constrained by semantic limitations but usually contain a focal event or an event-specific specificatory constituent. At the same time, these constructions only have the function of identifying their clausal heads. On the other hand, genitive relative constructions with semantic constraints are both descriptive at the semantic level and can distinguish or refer to their clausal heads at the pragmatic level.

Ultimately, this study argues that the acceptability of genitive relative constructions is not determined by syntax alone, but by the interaction between syntactic skeleton and semantic-pragmatic constraints on reference-point selection. I demonstrate that even syntactically well-formed strings (e.g. gapless relatives) can be unnatural when a possessee is highly individuated or lacks a cognitively salient dependency relation to a possessor, a finding that challenges purely structural accounts. By exploring these constraints (e.g. low salience, high accessibility), I offer a unified framework that explains both grammaticality and gradience in genitive relative constructions.

Bibliography

1. Bernardo, R. (1979). *The Function & Content Of Relative Clauses In Spontaneous Oral Narratives*. Proceedings of the 5th annual meeting of the Berkeley Linguistics Society, 5.
2. Carnesale, L. (2022). Predicative possessive constructions in Hindi. *Bhasha*, 1. https://doi.org/10.30687/bhasha/8409-3769/2022/01/009
3. Cheng, L. (1986). *De* In Mandarin. *Canadian Journal Of Linguistics*, 31(4), 313-326. https://doi.org/10.1017/S0008413100011865
4. Chi, X., Rao, G. & Xun, E. (2022). A quantitative study on monovalent noun and its ellipsis. In *Chinese Lexical Semantics* (pp. 388-398). Springer International Publishing. https://doi.org/10.1007/978-3-031-06547-7_30
5. Clark, E. V. (1978). Locationals: existential, locative, and possessive constructions. In Greenberg, J. H. (ed.), *Universals Of Human Language* (pp. 85-126). Stanford: Stanford University Press.
6. Cohen, E. (2019). Semitic genitive constructions: an expanded view. *Journal Of Semitic Studies*, 64(1), 1-50. https://doi.org/10.1093/jss/fgy032

7. Cohen, H. & Lefebvre, C. (2017). Handbook of categorization in cognitive science (2nd ed.). Elsevier.
8. Collins, C. (2007). *NYU Working Papers In Linguistics*, 1, 1-34.
9. Collins, C. & Radford, A. (2015). Gaps, ghosts and gapless relatives in spoken English. *Studia Linguistica*, 69 (2), 191-235.
10. Comrie, B. & Keenan, E. L. (1979). Noun phrase accessibility revisited. *Language*, 55(3), 649-664. https://doi.org/10.2307/413321
11. Croft, W. (1990). *Typology & Universals*. Cambridge University Press.
12. Dixon, R. M. W. (2010). *Basic Linguistic Theory*. Oxford University Press.
13. Du, J., Li, F. T., Ge, Y. & Zhang, J. (2023). Separation events in Mandarin, Russian and Korean: a crosslinguistic event-categorization study. *Review Of Cognitive Linguistics*, 21(2), 377-410. https://doi.org/10.1075/rcl.00143.du
14. Fox, B. A. (1981). Body part syntax: towards a universal characterization. *Studies In Language*, 3.
15. Fox, B. A. & Thompson, S. A. (1990). A discourse explanation of the grammar of relative clauses in English conversation. *Language*, 66(2), 297-316. https://doi.org/10.2307/414888
16. Gu, C. Y. (2001). 外界事物的“显著性”与句中名词的“有标性”——“出现、存在、消失”与“有界、无界”[The “salience” of external things and the “markedness” of nouns in sentences---“presence, existence, disappearance” and “boundedness, unboundedness”]. *现代语言学* [Contemporary Linguistics], 3(4), 264-274. https://doi.org/10.3969/j.issn.1007-8274.2001.04.003
17. Hu, X. H. (2009). 汉语话题结构制约的认知语义与语用分析 [A cognitive semantic and pragmatic account of the constraints on Chinese topic constructions]. *外国语* [Journal Of Foreign Language], 3, 60-68.
18. Joosten, F. (2010). Collective nouns, aggregate nouns, and superordinates: when “part of” and “kind of” meet. *Lingvisticæ Investigationes*, 1.
19. Ke, A. H. (2023). Syntax and semantics of NPs in Chinese possessive topic constructions. *Journal Of East Asian Linguistics*, 32(2), 133-167. https://doi.org/10.1007/s10831-023-09254-w
20. Kou, X. & Yuan, Y. L. (2017). 汉语定语小句的类型及其句法表现 [The types and syntactic representation of Chinese attributive clauses]. *外语教学与研究* [Language Teaching & Research], 4, 67-78.

21. Kou, X. & Yuan, Y. L. (2019). 功用义对关系小句结构类型和解读倾向的影响——从现代汉语工具、材料、处所成分关系化的限制条件谈起 [The influence of functional meaning on the structural types and interpretive tendencies of relative clauses: starting from the constraints of relativizing tools, materials, and locations in modern Chinese]. *中国语文* [Chinese Language], 6.
22. Langacker, R. W. (2013). *Essentials Of Cognitive Grammar* (1st ed.). Oxford University Press.
23. Langacker, R. W. (1993). Reference-point constructions. *Cognitive Linguistics*, 4(1), 1-38. https://doi.org/10.1515/cogl.1993.4.1.1
24. Lee, J. & Lee, C. (2012). *Gap In "Gapless" Relative Clauses In Korean & Other Asian Languages*. UCLA working papers in linguistics. Theories of everything, 17, 204-214.
25. Li, C. & Kang, S.-M. (2024). Gaplessness in Chinese relative clauses. *Lingua*, 312, 103828-. https://doi.org/10.1016/j.lingua.2024.103828
26. Liao, Q. Z. (1986). 篇章中的框-梴关系与所指的确定 [Frame-pane relations and referential identification in Chinese]. *语法研究和探索* [Grammar Research & Exploration]. Peking University Press.
27. Lieder, F. & Griffiths, T. L. (2020). Resource-rational analysis: understanding human cognition as the optimal use of limited computational resources. *The Behavioral & Brain Sciences*, 43, e1. https://doi.org/10.1017/S0140525X1900061X
28. Lin, C.-J. C. (2018). Subject prominence and processing dependencies in prenominal relative clauses: the comprehension of possessive relative clauses and adjunct relative clauses in Mandarin Chinese. *Language*, 94(4), 758-797. https://doi.org/10.1353/lan.2018.0053
29. Liu, D. Q. (2002). 汉语类指成分的语义属性和句法属性 [Semantic and pragmatic attributes of Chinese generic reference]. *中国语文* [Chinese Language], 5.
30. Loock, R. (2010). Appositive Relative Clauses In English: Discourse Functions & Competing Structures (1st ed.). John Benjamins.
31. Lu, B. F. (2003).“的”的基本功能和派生功能——从描写性到区别性再到指称性 [The basic and derivative functions of *de* in Chinese: from descriptiveness to differentiability and referentiality]. *世界汉语教学* [World Chinese Teaching], 1.
32. Lu, J. M. (2002). 再谈“吃了他三个苹果”一类结构的性质 [“Ate 3 apples of him”: reanalysis of its structure]. *中国语文* [Studies Of The Chinese Language], 4.
33. Lyons, J. (1967). A note on possessive, existential and locative sentences. *Foundations Of*

Language, 3(4), 390-396.

34. Murasugi, K. (1991). *Noun Phrases In Japanese & English: A Study In Syntax, Learnability & Acquisition*. PhD dissertation: University Of Connecticut.
35. Ning, C. Y. (1993). *The Overt Syntax Of Relativization & Topicalization In Chinese*. PhD dissertation: University Of California, Irvine
36. Öztürk, B. & Taylan, E. E. (2016). Possessive constructions in Turkish. *Lingua*, 182, 88-108. https://doi.org/10.1016/j.lingua.2015.08.008
37. Pan, H. (2022). Remarks on gapless relative clauses and complement clauses in Mandarin Chinese. *Asian Languages & Linguistics*, 3(1), 109-140. https://doi.org/10.1075/alal.21008.pan
38. Paul, W. (2017). The insubordinate subordinator *de* in Mandarin Chinese: second take. In Tang, Sze-Wing (ed.). *汉语"的"的研究* [Studies Of The Particle *De* In Chinese]. Frontiers In Chinese Linguistics Series. Peking University Press.
39. Shi, C. H. (2002). 试析名词的语义结构 [Analysis of the semantic structure of nouns]. *世界汉语教学* [World Chinese Teaching], 4, 17-25. https://doi.org/10.3969/j.issn.1002-5804.2002.04.003
40. Si, F. Z. (2017). Two types of Chinese possessive structures. In *Studies On Syntactic Cartography*. China Social Sciences Press.
41. Tang, Z. D. (2007). 汉语主句现象进入关系从句初探 [A preliminary study of Chinese main clauses entering relative clauses]. *语法研究和探索* [Grammar Research & Exploration]. Beijing: The Commercial Press.
42. Tao, H. Y. (2002). 汉语口语叙事体关系从句结构的语义和篇章属性 [The semantic and discoursal attributes of relative clauses in spoken Chinese in a narrative style]. *现代中国语研究* [Modern Chinese Research], 4.
43. Taylor, J. R. (2003). *Linguistic Categorization* (3rd ed.). Oxford University Press. https://doi.org/10.1093/oso/9780199266647.001.0001
44. Truswell, R. (2011). *Events*. Oxford University Press.
45. Tsai, H.-C.-J. (2008). On gapless relative clauses in Chinese. *Nanzan Linguistics*, Special Issue 5, 109-124.
46. Ünal, E., Ji, Y. & Papafragou, A. (2021). From event representation to linguistic meaning. Topics In Cognitive Science, 13(1), 224-242. https://doi.org/10.1111/tops.12475
47. Wan, Q. (2012). 超越区别与描写之争: "的"的认知入场作用 [Transcending the argument

between distinction and description: the cognitive role of *de* in Chinese]. *世界汉语教学* [World Chinese Teaching], 2.

48. Wang, J. (2005). 体词的陈述性与非个体性 [Declarativeness and non-individuality of aspectual words]. *Journal of East China Normal University*, 6.
49. Wang, M. & Zhao, H. (2024). Are schematic diagrams valid visual representations of concepts? Evidence from mental imagery in online processing of English prepositions. *Language & Cognition*, 16(4), 1564-1587. https://doi.org/10.1017/langcog.2024.18
50. Wen, H. Y. (2020). *Relative Clauses In Mandarin Chinese*. PhD dissertation: Queen Mary University Of London
51. Winters, M. E. & Nathan, G. S. (2020). *Cognitive Linguistics For Linguists* (1st ed.). Springer International Publishing.
52. Yuan, Y. L. (1994). 一价名词的认知研究 [Cognitive research on monovalent nouns]. *中国语文* [Chinese Language], 4, 241-253.
53. Zhang, B. J. (2014). 汉语句法结构的观察角度 [The analytical angle of Chinese syntactic structures]. *语法研究和探索* [Grammar Research & Exploration]. Beijing: The Commercial Press.
54. Zhang, B. (2018). 现代汉语的非论元性句法成分 [Non-argument syntactic constituents in modern Chinese]. *World Chinese Teaching*, 4, 442-.
55. Zhang, X. J. & Wang, F. F. (2025). 论现代汉语“N1 的 N2”结构的语义类型 [On the semantic types of the “N2 of N1” structure in Modern Chinese]. *台州学院学报* [Journal of Taizhou University], 47(5), 58-65.
56. Zhu, D. X. (1961). 说”的” [About *De*]. *中国语文* [Chinese Language]. 12.
57. Zhu, D. X. (1978). “的”字结构和判断句 [The structure related to *De* and declarative sentences]. *中国语文* [Chinese Language]. 1-2.

Supplementary Material

Please see the critical stimuli via https://docs.google.com/document/d/1VV49yyfcbkTdE0CNs8EUD-3YCSVWbxuzA-rXmk5RdWs/edit?usp=drive_web&ouid=105506636227007263165.